

Deep Learning – Practice Paper 2

“Computation Grinder”

24 Questions — 52 Marks — Suggested Time: 2.5 hours

Heavy on numerical SA problems — Difficulty: Moderate–Hard

IITM BS Pattern

Type	Count	Marks
MCQ	6	14
MSQ	4	10
SA	14	28
Total	24	52

Q1 [SA – 2 marks]

A $7 \times 7 \times 3$ image. Conv with 12 filters of size 5×5 , stride 1, padding 2, **with bias**. How many learnable parameters?

Q2 [SA – 2 marks]

Compute convolution (stride 1, no padding):

$$X = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 3 & 1 & 0 & 2 \\ 0 & 2 & 1 & 1 \\ 1 & 0 & 3 & 0 \end{bmatrix}, \quad W = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Give the element at position $(0, 0)$ of the output.

Q3 [SA – 2 marks]

A CNN pipeline: Input $10 \times 10 \times 1$. Conv(3×3 , 8 filters, stride 1, padding 0, bias) $\rightarrow$ ReLU $\rightarrow$ MaxPool(2×2 , stride 2) $\rightarrow$ Conv(3×3 , 16 filters, stride 1, padding 0, bias). What is the final output dimension?

Q4 [SA – 3 marks]

CBOV model. Vocabulary = {sun(0), moon(1), star(2), sky(3)}.

$$C = \begin{bmatrix} 2 & 0 & -1 & 1 \\ 1 & -1 & 2 & 0 \end{bmatrix}, \quad W = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \\ 1 & 1 \end{bmatrix}$$

Context words = {sun, star} (window $d = 1$). Compute $P(\text{sky}|\text{sun, star})$. Round to 2 decimal places.

Q5 [SA – 3 marks]

Skip-gram, window $d = 3$ (each side). Sentence: “neural networks learn complex representations from data” (7 words). How many unique (target, context) pairs?

Q6 [SA – 3 marks]

Co-occurrence matrix. Corpus (stop words removed):

1. “cats chase mice” 2. “dogs chase cats” 3. “mice fear dogs”

Window = 1. Compute $X[\text{chase, cats}]$ and determine the sign of $\text{PMI}(\text{chase, cats})$.

Q7 [SA – 2 marks]

RNN: $s_t = \sigma(Ux_t + Ws_{t-1})$, $U = 1$, $W = 2$, $s_0 = 0$.

Input $x_1 = -\ln 3$. Find s_1 .

Then input $x_2 = 0$. Find s_2 . (*Hint: $\sigma(0.5) \approx 0.622$.*)

Q8 [SA – 2 marks]

From Q7, $V = 4$, linear output. Compute $\hat{y}_2 = V \cdot s_2$. Then compute MSE loss $L = \frac{1}{2}(\hat{y}_2 - y)^2$ with $y = 3$. Round to 2 decimal places.

Q9 [SA – 2 marks]

LSTM: $x_t \in \mathbb{R}^6$, $h_t \in \mathbb{R}^8$, $\hat{y}_t \in \mathbb{R}^5$, $T = 20$, with bias. Total trainable parameters?

Q10 [SA – 2 marks]

GRU: same dimensions as Q9 ($n = 6$, $d = 8$, $k = 5$), with bias. Total trainable parameters?

Q11 [SA – 3 marks]

Attention score: $e_{jt} = V_{\text{att}}^T \tanh(U_{\text{att}} \cdot s_{t-1} + W_{\text{att}} \cdot h_j)$.

$h_1 = [2, 0]^T$, $h_2 = [-1, 1]^T$, $h_3 = [0, 3]^T$, $s_{t-1} = [1, -1]^T$.

$U_{\text{att}} = W_{\text{att}} = I$, $V_{\text{att}} = [1, -1]^T$.

Compute e_1 , e_2 , e_3 and find which encoder state gets the highest attention weight.

Q12 [SA – 2 marks]

Transformer: $d_{\text{model}} = 128$, $h = 4$, $d_k = d_v = 32$, no bias. Total MHA parameters?

Q13 [MCQ – 2 marks]

ResNet: $f(x) = x + \text{ReLU}(w \cdot x)$, two blocks. $w_1 = -2$, $w_2 = 3$, input $x = 2$. Final output?

- A. 2
- B. 4
- C. 8
- D. 10

Q14 [SA – 2 marks]

Attention scores: $B = 6$ batches, $T = 16$ sequence length, $h = 4$ heads. How many total attention score values are computed?

Q15 [MCQ – 2 marks]

Match the attention mechanism:

(P) Multi-Head Attention (Encoder) (Q) Masked Self-Attention (R) Cross Attention (S)

FFN

(1) Queries from decoder, K/V from encoder (2) Prevents future token access (3) Source attends to all source (4) Position-wise after attention

- A. P→2, Q→3, R→1, S→4
- B. P→3, Q→1, R→2, S→4
- C. P→3, Q→2, R→1, S→4
- D. P→4, Q→2, R→3, S→1

Q16 [MCQ – 2 marks]

For the Adam optimizer, which is the bias-corrected first moment estimate?

- A. $\hat{m}_t = m_t / (1 - \beta_1^t)$
- B. $\hat{m}_t = m_t / (1 - \beta_2^t)$
- C. $\hat{m}_t = v_t / (1 - \beta_1^t)$
- D. $\hat{m}_t = m_t \cdot \beta_1^t$

Q17 [MSQ – 3 marks]

Which methods address vanishing gradients? (Select all correct.)

- A. Replace sigmoid with ReLU
- B. He/Xavier initialization
- C. Skip connections
- D. Increase training data
- E. LSTM/GRU

Q18 [MCQ – 2 marks]

CBOW: vocabulary size $|V| = 10000$, embedding dimension $k = 300$. How many total parameters in the model (context matrix C and word matrix W)?

- A. 3,000,000
- B. 6,000,000
- C. 300,000
- D. 10,000

Q19 [MSQ – 2 marks]

Which activation functions can cause the “dead neuron” problem?

- A. Sigmoid
- B. tanh
- C. ReLU
- D. Leaky ReLU

Q20 [MCQ – 2 marks]

In mini-batch gradient descent with $N = 1000$ data points and batch size $B = 50$, how many parameter updates (steps) occur in one epoch?

- A. 1
- B. 20
- C. 50
- D. 1000

Q21 [MSQ – 3 marks]

For the sigmoid function $\sigma(x) = 1/(1 + e^{-x})$, which are true?

- A. $\sigma(0) = 0.5$
- B. $\sigma'(0) = 0.25$
- C. $\sigma'(x) = \sigma(x)(1 - \sigma(x))$
- D. $\sigma(x)$ is zero-centered
- E. Maximum derivative occurs at $x = 0$

Q22 [MSQ – 2 marks]

A co-occurrence matrix has which of these properties?

- A. It is symmetric (if word and context sets are the same)
- B. Its dimensions grow with vocabulary size
- C. It is typically dense
- D. One-hot vectors have cosine similarity 0

Q23 [MCQ – 2 marks]

In a Transformer encoder, the computation is parallelized:

- A. Along the vertical direction (across layers)
- B. Along the horizontal direction (across positions within a layer)
- C. Along both directions
- D. Neither direction

Q24 [SA – 2 marks]

FFN: $d_{\text{model}} = 512$, $d_{\text{ff}} = 2048$, with bias. Total FFN parameters?

SOLUTIONS – Practice Paper 2

Q1: $5 \times 5 \times 3 \times 12 + 12 = 900 + 12 = \mathbf{912}$.

Q2: Position $(0, 0)$ uses top-left 3×3 of X : $\begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$.

Element-wise with W : $1(1)+0(0)+2(-1)+3(0)+1(1)+0(0)+0(-1)+2(0)+1(1) = 1-2+1+1 = \mathbf{1}$.

Q3: Conv1: $1 + (10 - 3)/1 = 8$. Out: $8 \times 8 \times 8$. Pool: $8/2 = 4$. Out: $4 \times 4 \times 8$. Conv2: $1 + (4 - 3)/1 = 2$. Final: $\mathbf{2 \times 2 \times 16}$.

Q4: $u_{\text{sun}} = C[:, 0] = [2, 1]^T$, $u_{\text{star}} = C[:, 2] = [-1, 2]^T$. $h = [2, 1] + [-1, 2] = [1, 3]^T$.

$$z = Wh = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -3 \\ 4 \end{bmatrix}.$$

$e^1 = 2.718$, $e^2 = 7.389$, $e^{-3} = 0.050$, $e^4 = 54.598$. Sum = 64.755.

$P(\text{sky}) = 54.598/64.755 = \mathbf{0.84}$.

Q5: 7 words, $d = 3$: $0 + 3 = 3$, $1 + 3 = 4$, $2 + 3 = 5$, $3 + 3 = 6$, $3 + 2 = 5$, $3 + 1 = 4$, $3 + 0 = 3$. Total = **30**.

Q6: Pairs (window=1): S1: cats–chase, chase–mice. S2: dogs–chase, chase–cats. S3: mice–fear, fear–dogs.

$X[\text{chase, cats}] = \mathbf{2}$ (S1: cats–chase, S2: chase–cats).

Row sums: cats=3, chase=4, mice=2, dogs=2, fear=2. $N = 13$? Let me recount. Co-occurrence matrix (symmetric): cats-chase:2, chase-mice:1, dogs-chase:1, mice-fear:1, fear-dogs:1. All entries sum: $2(2 + 1 + 1 + 1 + 1) = 12$. So $N = 12$.

count(chase)=4, count(cats)=3 (from rows). Actually let me be careful. The symmetric matrix entries: chase row: cats=2, mice=1, dogs=1 $\rightarrow$ sum=4. cats row: chase=2, dogs=1 $\rightarrow$ sum=3.

PMI = $\ln(2 \times 12 / (4 \times 3)) = \ln(24/12) = \ln 2 \approx 0.693$. **Positive**.

Q7: $s_1 = \sigma(-\ln 3 + 0) = 1/(1 + 3) = 0.25$. $s_2 = \sigma(0 + 2(0.25)) = \sigma(0.5) \approx \mathbf{0.622}$.

Q8: $\hat{y}_2 = 4 \times 0.622 = 2.488$. $L = \frac{1}{2}(2.488 - 3)^2 = \frac{1}{2}(0.262) = \mathbf{0.13}$.

Q9: $n = 6, d = 8, k = 5$. $4d(n + d + 1) + k(d + 1) = 4(8)(15) + 5(9) = 480 + 45 = \mathbf{525}$. T irrelevant.

Q10: GRU: $3d(n + d + 1) + k(d + 1) = 3(8)(15) + 5(9) = 360 + 45 = \mathbf{405}$.

Q11: $e_1: s + h_1 = [1, -1] + [2, 0] = [3, -1]$. $\tanh = [0.995, -0.762]$. $e_1 = [1, -1] \cdot [0.995, -0.762] = 0.995 + 0.762 = 1.757$.

$e_2: [1, -1] + [-1, 1] = [0, 0]$. $\tanh = [0, 0]$. $e_2 = 0$.

$e_3: [1, -1] + [0, 3] = [1, 2]$. $\tanh = [0.762, 0.964]$. $e_3 = [1, -1] \cdot [0.762, 0.964] = 0.762 - 0.964 = -0.202$.

Highest score: $e_1 = 1.757$. h_1 gets highest attention.

Q12: $d_k = 128/4 = 32$. Per head: $(32 + 32 + 32) \times 128 = 12288$. 4 heads: 49152. W_O : $128 \times 128 = 16384$. Total = **65536**.

Q13: C. Block 1: $2 + \text{ReLU}(-4) = 2 + 0 = 2$. Block 2: $2 + \text{ReLU}(6) = 2 + 6 = 8$.

Q14: $6 \times 4 \times 16 \times 16 = \mathbf{6144}$.

Q15: C. $P \rightarrow 3$, $Q \rightarrow 2$, $R \rightarrow 1$, $S \rightarrow 4$.

Q16: A. $\hat{m}_t = m_t / (1 - \beta_1^t)$.

Q17: A, B, C, E. Increasing data doesn't fix vanishing gradients.

Q18: B. $C: k \times |V| = 3M$, $W: |V| \times k = 3M$. Total = $6M$.

Q19: C. ReLU only. Leaky ReLU was designed to avoid it. Sigmoid/tanh saturate but don't "die."

Q20: B. $N/B = 1000/50 = 20$ updates per epoch.

Q21: A, B, C, E. D is false—sigmoid outputs $(0, 1)$, not zero-centered.

Q22: A, B, D. C is false—co-occurrence matrices are typically sparse.

Q23: B. Parallelized horizontally (across positions), not vertically (across layers).

Q24: $2(512)(2048) + 2048 + 512 = 2097152 + 2560 = \mathbf{2099712}$.